Annex I. Updates on Debt Sustainability Analysis[1]

This Annex provides an update to the comprehensive Debt Sustainability Analysis (DSA) conducted in the context of the May 2016 Article IV consultation and Fourth Review. The fiscal adjustment under the Homegrown Program and strong economic recovery have led to a further improvement in the projected debt trajectory. However, the overall assessment is unchanged, and Grenada's external debt remains "in distress" until all arrears to official creditors are regularized.[2]

1. Grenada's external debt remains in distress as debt restructuring is not yet completed. The baseline scenario in this Debt Sustainability Analysis (DSA) update incorporates all debt restructuring agreements executed thus far, including the agreed second-step haircut on the new commercial bonds ("2030 bonds") due in 2017 after successful completion of the ECF-supported program. In this scenario, Grenada's public debt is projected to decline to 84.3 percent of GDP at end-2016 before gradually trending down toward 57.5 percent of GDP by 2020, below the regional debt target of 60 percent of GDP. Meanwhile, public external debt is projected to decline from 62.6 percent of GDP in 2015 to 40.8 percent of GDP by 2020. Nevertheless, the DSA update finds that the debt thresholds on the present value of the debt-to-GDP, debt service-to-revenue, and debt-to-exports ratios are breached. These breaches are temporary in nature, as the amortization of the 2030 bonds leads to a rapid decline in the present value of external debt. However, Grenada's external debt would remain "in distress" due to the remaining unresolved arrears to official creditors.[3] (Grenada also continues to have membership arrears to international organizations.) Public domestic debt, at 30 percent of GDP at end-2015, remains a relatively smaller share of the total debt burden, but related vulnerabilities need to be monitored closely.

2. The macroeconomic outlook has improved, but long term fiscal prudence is necessary to restore debt sustainability. Compared to the previous DSA, growth surprised on the upside in 2015—an increase of 6.2 percent relative to the initial estimate of 4.6 percent—and is projected to maintain its momentum in 2016 before converging to its potential in the medium-term. An upward revision to oil price forecasts translates into a higher current account deficit in the medium term. (see text table below). Meanwhile, fiscal consolidation advanced faster and stronger than envisaged by the program, which has contributed to accelerating debt reduction. Going forward, a strong and credible medium term fiscal framework, which underpins the baseline scenario in the DSA, is critical for ensuring the long-term sustainability of Grenada's public debt by putting it on a firm downward trend. In particular, the second-step haircut on the 2030 bonds (about 8 percent of GDP) hinges on adherence to the framework to ensure a successful completion of the ECF-supported adjustment program. And the projected primary surplus above 3.5 percent of GDP through 2020 marks a break from the past and rests on the assumption that the FRA is strictly adhered to. The historical scenario

[1] Prepared by IMF and World Bank staffs.

[2] The Country Policy and Institutions Assessment (CPIA) ranks Grenada as a medium performer (the average CPIA rating in 2013-15 is 3.52).

[3] See staff report ¶6 and MEFP ¶6 for details on the debt restructuring process.

(Figure AI:2) illustrates how a return to lax fiscal policies would put Grenada's public debt back on an unsustainable path. The experience of the past decade, with moderate GDP growth, also highlights the tourism sector's exposure to macroeconomic conditions in advanced economies and to natural disaster risk. A customized shock scenario (Figure AI:2) analyzes the impact of the latter. Further risks to the outlook are posed by shocks to imports, particularly fuel, prices.

DSA Update: Macroeconomic Assumptions
(In percent of GDP, unless otherwise specified)

	Historical Average	Projections 2016-21		Projections 2022-36	
		4th Review	5th Review	4th Review	5th Review
Non-interest current account balance	-20.3	-12.9	-14.4	-14.2	-14.4
Real GDP growth (in percent)	1.2	2.7	2.8	2.7	2.7
Growth of exports of G&S (USD terms, in percent)	3.0	5.9	5.8	5.8	5.8
Current official transfer	1.4	0.6	1.0	0.4	0.8
Net FDI	9.3	8.1	8.8	8.7	9.1
Primary balance	-1.9	3.4	3.7	-0.7	-1.0
Revenue and grants	23.3	25.3	24.5	25.5	23.2
of which: grants	3.0	3.1	2.8	3.1	2.2
Primary (noninterest) expenditure	25.1	21.9	20.7	26.2	24.3
Inflation rate (GDP deflator, in percent)	1.4	2.3	2.3	2.3	2.3
Memorandum item:		4th Review	5th Review		
2015 Nominal GDP (in million USD)		953.6	984.1		

Source: Grenadian authorities and IMF staff projections

3. Renewed efforts are needed to swiftly conclude the comprehensive debt restructuring. Total debt reduction directly attributed to restructuring savings thus far amounts to 17.1 percent of projected 2016 GDP. One half of that was already executed and the rest is expected to be granted in 2017. While payments on restructured debt instruments have been restored (including on the 2030 bonds), discussions have recently started or been intensified with the remaining more dispersed groups of creditors (see MEFP, paragraph 6). Three and half years after default, it is imperative that Grenada normalize relations with creditors by clearing or regularizing arrears, an important step toward lowering the risk of debt distress. The authorities reiterated their commitment to concluding the debt restructuring by the end of the ECF-supported program in 2017 and to adhering to the fiscal responsibility law.

Table A1.1. Grenada: Public Debt Sustainability Framework, 2013-36

(In percent of GDP)

	Actual					Estimate	Projections								
	2013	2014	2015	Average 5/	Standard Deviation 5/	2016	2017	2018	2019	2020	2021	2016-21 Average	2026	2036	2022-36 Average
Public sector debt 1/	108.1	101.8	91.7			84.3	72.9	67.2	62.3	57.6	54.2		54.4	54.2	
of which: foreign-currency denominated	70.8	67.7	62.6			59.9	51.9	47.4	44.1	40.6	38.0		37.7	37.0	
Change in public sector debt	4.7	-6.3	-10.0			-7.4	-11.4	-5.8	-4.9	-4.7	-3.4		0.0	0.0	
Identified debt-creating flows	2.0	-5.4	-12.7			-6.7	-11.4	-5.8	-5.0	-4.7	-3.4		-0.2	-0.3	
Primary deficit	3.9	1.1	-2.2	1.9	1.9	-4.8	-4.1	-4.0	-3.8	-3.5	-2.3	-3.7	1.0	1.0	1.0
Revenue and grants	20.9	24.5	24.8			25.1	24.6	24.5	24.3	24.2	24.1		23.5	22.6	
of which: grants	1.4	4.1	3.3			3.0	2.9	2.8	2.7	2.7	2.6		2.3	1.9	
Primary (noninterest) expenditure	24.8	25.6	22.6			20.3	20.5	20.5	20.5	20.7	21.8		24.5	23.6	
Automatic debt dynamics	-1.9	-4.6	-4.1			-0.9	-2.0	-1.4	-1.2	-1.2	-1.1		-1.2	-1.3	
Contribution from interest rate/growth differential	-1.1	-5.3	-3.8			-0.8	-1.3	-1.3	-1.1	-1.1	-1.0		-1.1	-1.1	
of which: contribution from average real interest rate	1.3	2.1	2.2			2.0	1.0	0.6	0.6	0.6	0.5		0.4	0.3	
of which: contribution from real GDP growth	-2.4	-7.4	-6.0			-2.7	-2.3	-1.9	-1.8	-1.6	-1.5		-1.4	-1.4	
Contribution from real exchange rate depreciation	-0.8	0.7	-0.3			-0.2	-0.7	0.0	0.0	-0.1	-0.1		...	...	
Other identified debt-creating flows	0.0	-1.9	-6.4			-0.9	-5.3	-0.5	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)6/	0.0	-1.9	-5.8			-1.0	-5.3	-0.5	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	-0.6			0.1	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.7	-0.9	2.7			-0.8	0.0	0.0	0.1	0.0	0.0		0.2	0.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	77.8			70.0	59.0	53.9	49.5	45.4	42.4		41.4	43.7	
of which: foreign-currency denominated	...	...	48.7			45.6	38.0	34.1	31.3	28.4	26.2		24.8	26.5	
of which: external	...	...	48.7			45.6	38.0	34.1	31.3	28.4	26.2		24.8	26.5	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	7.3	22.3	17.3			16.7	13.4	12.1	11.7	11.3	11.9		12.0	8.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	314.2			278.5	240.0	220.2	203.3	187.4	175.9		176.1	193.4	
PV of public sector debt-to-revenue ratio (in percent)	...	...	362.2			315.8	271.8	248.8	229.1	210.8	197.4		195.6	211.0	
of which: external 3/	...	...	226.8			205.6	174.9	157.5	145.2	131.9	122.1		117.0	128.0	
Debt service-to-revenue and grants ratio (in percent) 4/	15.9	86.6	78.5			85.3	71.3	65.7	63.6	61.1	58.8		46.7	33.7	
Debt service-to-revenue ratio (in percent) 4/	17.0	103.9	90.5			96.7	80.7	74.3	71.7	68.8	66.0		51.9	36.8	
Primary deficit that stabilizes the debt-to-GDP ratio	-0.8	7.4	7.8			2.6	7.3	1.8	1.1	1.2	1.1		1.0	1.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	2.4	7.3	6.2	1.2	4.7	3.1	2.7	2.7	2.7	2.7	2.7	2.8	2.7	2.7	2.7
Average nominal interest rate on forex debt (in percent)	2.6	3.0	3.7	2.5	0.7	3.3	3.1	2.7	2.6	2.5	2.5	2.8	2.0	1.8	1.9
Average real interest rate on domestic debt (in percent)	1.9	3.9	1.8	2.9	1.3	2.7	1.2	1.9	2.0	1.9	1.9	1.9	2.0	2.1	2.0
Real exchange rate depreciation (in percent, + indicates depreciation)	-1.3	1.0	-0.5	0.1	1.4	-0.3	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	2.9	0.8	1.6	1.4	1.5	1.5	3.2	2.3	2.2	2.3	2.3	2.3	2.3	2.3	2.3
Growth of real primary spending (deflated by GDP deflator, in percent)	9.1	10.8	-6.2	1.9	6.0	-7.3	3.6	2.7	2.9	3.6	8.1	2.3	2.3	2.4	3.3
Grant element of new external borrowing (in percent)	...	...	...	...	...	29.1	13.8	33.5	32.9	32.6	29.7	28.6	29.7	29.7	29.7

Sources: Country authorities; and staff estimates and projections.

1/ Comprises central government and government guaranteed debt

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

6/Debt relief includes savings from restructuring agreements reached with external and domestic creditors.

Table A1.2. Grenada: External Debt Sustainability Framework, 2013-36

(In percent of GDP)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2013	2014	2015			2016	2017	2018	2019	2020	2021	2016-2021 Average	2026	2036	2022-203 Average
External debt (nominal) 1/	**144.7**	**144.8**	**137.1**			**132.8**	**122.5**	**116.8**	**112.5**	**107.9**	**104.3**		**98.8**	**88.6**	
of which: public and publicly guaranteed (PPG)	70.8	67.7	62.6			59.9	51.9	47.4	44.1	40.6	38.0		37.7	37.0	
Change in external debt	3.7	0.1	-7.7			-4.3	-10.3	-5.7	-4.3	-4.6	-3.6		-1.1	-0.9	
Identified net debt-creating flows	2.6	1.2	-4.0			2.1	3.2	3.9	4.1	4.5	5.6		4.0	3.4	
Non-interest current account deficit	**21.4**	**14.4**	**14.1**	**20.3**	**4.3**	**12.4**	**13.3**	**14.2**	**14.8**	**15.3**	**16.4**		**14.9**	**14.1**	14.6
Deficit in balance of goods and services	21.9	15.6	14.8			12.5	14.1	15.0	15.7	16.1	17.2		15.2	13.8	
Exports	28.5	28.0	27.2			28.2	27.8	27.9	28.0	28.1	28.1		28.8	31.0	
Imports	50.4	43.6	42.0			40.7	41.9	42.9	43.7	44.2	45.4		44.0	44.7	
Net current transfers (negative = inflow)	-2.2	-2.9	-2.2	-3.5	1.0	-2.7	-2.6	-2.5	-2.6	-2.6	-2.6		-2.3	-1.8	-2.1
of which: official	-0.2	-1.1	-0.6			-1.1	-1.1	-1.0	-1.0	-1.0	-1.0		-0.9	-0.7	
Other current account flows (negative = net inflow)	1.7	1.7	1.5			2.5	1.8	1.7	1.7	1.7	1.7		2.0	2.1	
Net FDI (negative = inflow)	**-13.4**	**-4.4**	**-9.9**	**-9.3**	**4.7**	**-8.3**	**-8.6**	**-8.6**	**-9.1**	**-9.1**	**-9.1**		**-9.1**	**-9.1**	-9.1
Endogenous debt dynamics 2/	**-5.3**	**-8.8**	**-8.2**			**-1.9**	**-1.5**	**-1.6**	**-1.6**	**-1.6**	**-1.6**		**-1.8**	**-1.6**	
Contribution from nominal interest rate	1.9	2.1	2.5			2.1	1.9	1.5	1.4	1.3	1.1		0.8	0.7	
Contribution from real GDP growth	-3.1	-9.8	-8.4			-4.0	-3.4	-3.1	-3.0	-2.9	-2.8		-2.6	-2.3	
Contribution from price and exchange rate changes	-4.0	-1.1	-2.3			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**1.1**	**-1.1**	**-3.6**			**-6.5**	**-13.5**	**-9.6**	**-8.4**	**-9.1**	**-9.2**		**-5.1**	**-4.3**	
of which: exceptional financing	-0.7	-3.4	-3.1			-0.5	-2.1	0.0	0.0	0.0	0.0		0.0	0.0	
debt restructuring	0.0	-1.9	-4.9			-0.6	-4.5	-0.5	0.0	0.0	0.0				
capital transfers						-6.1	-6.7	-6.4	-6.8	-6.7	-6.7		-6.3	-5.7	
PV of external debt 4/	...	...	123.2			118.5	108.6	103.6	99.7	95.7	92.5		85.8	78.1	
In percent of exports	...	...	452.2			420.4	390.4	371.7	356.5	341.1	328.7		297.8	252.2	
PV of PPG external debt	**...**	**...**	**48.7**			**45.6**	**38.0**	**34.1**	**31.3**	**28.4**	**26.2**		**24.8**	**26.5**	
In percent of exports	**...**	**...**	**178.9**			**161.7**	**136.6**	**122.5**	**112.0**	**101.2**	**93.2**		**86.0**	**85.5**	
In percent of government revenues	**...**	**...**	**226.8**			**205.6**	**174.9**	**157.5**	**145.2**	**131.9**	**122.1**		**117.0**	**128.0**	
Debt service-to-exports ratio (in percent)	**6.5**	**13.2**	**16.7**			**19.1**	**17.2**	**16.4**	**14.7**	**14.4**	**13.4**		**9.5**	**7.3**	
PPG debt service-to-exports ratio (in percent)	**5.9**	**12.6**	**16.0**			**18.5**	**16.6**	**15.7**	**14.1**	**13.7**	**12.8**		**9.0**	**7.0**	
PPG debt service-to-revenue ratio (in percent)	**8.6**	**17.2**	**20.3**			**23.5**	**21.2**	**20.2**	**18.2**	**17.9**	**16.8**		**12.2**	**10.5**	
Total gross financing need (Millions of U.S. dollars)	82.4	125.1	85.8			96.8	103.7	116.2	118.1	128.3	145.8		143.2	202.0	
Non-interest current account deficit that stabilizes debt ratio	17.6	14.3	21.7			16.7	23.6	19.9	19.2	19.9	20.0		16.0	15.0	
Key macroeconomic assumptions															
Real GDP growth (in percent)	2.4	7.3	6.2	1.2	4.7	3.1	2.7	2.7	2.7	2.7	2.7	2.8	2.7	2.7	2.7
GDP deflator in US dollar terms (change in percent)	2.9	0.8	1.6	1.4	1.5	1.5	3.2	2.3	2.2	2.3	2.3	2.3	2.3	2.3	2.3
Effective interest rate (percent) 5/	1.4	1.6	1.9	1.5	0.2	1.6	1.6	1.3	1.2	1.2	1.1	1.3	0.9	0.8	0.9
Growth of exports of G&S (US dollar terms, in percent)	3.8	6.3	5.1	3.0	6.2	8.2	4.7	5.3	5.4	5.3	5.3	5.7	5.6	5.2	5.7
Growth of imports of G&S (US dollar terms, in percent)	7.2	-6.4	4.1	-0.8	9.6	1.3	9.3	7.4	6.9	6.3	7.8	6.5	5.2	5.2	5.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	29.1	13.8	33.5	32.9	32.6	29.7	28.6	29.7	29.7	29.7
Government revenues (excluding grants, in percent of GDP)	19.5	20.4	21.5			22.2	21.7	21.7	21.6	21.5	21.5		21.2	20.7	21.0
Aid flows (in Millions of US dollars) 7/	11.6	37.2	32.3			31.1	51.2	40.1	40.9	41.8	34.9		39.9	52.3	
of which: Grants	11.6	37.2	32.3			30.6	31.4	32.2	33.1	34.0	34.9		39.9	52.3	
of which: Concessional loans	0.0	0.0	0.0			0.6	19.8	7.8	7.8	7.8	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			4.2	3.1	3.3	3.3	3.1	3.2		3.4	2.9	3.2
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			58.4	72.2	76.8	74.3	77.4	69.6		57.4	55.3	57.2
Memorandum items:															
Nominal GDP (Millions of US dollars)	842.6	911.6	984.1			1029.8	1091.5	1147.2	1204.1	1264.5	1327.9		1697.8	2783.6	
Nominal dollar GDP growth	5.3	8.2	8.0			4.6	6.0	5.1	5.0	5.0	5.0	5.1	5.1	5.1	5.1
PV of PPG external debt (in Millions of US dollars)			479.7			469.1	414.8	391.5	377.5	359.2	348.0		420.5	737.3	
(PVt-PVt-1)/GDPt-1 (in percent)						-1.1	-5.3	-2.1	-1.2	-1.5	-0.9	-2.0	1.2	1.5	1.3
Gross workers' remittances (Millions of US dollars)	30.5	31.2	31.8			32.7	33.9	35.4	39.8	44.3	48.9		59.4	87.6	
PV of PPG external debt (in percent of GDP + remittances)	...	...	47.2			44.2	36.9	33.1	30.3	27.4	25.3		23.9	25.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	159.9			145.3	122.9	110.3	100.2	90.0	82.4		76.6	77.7	
Debt service of PPG external debt (in percent of exports + remittanc	...	...	14.3			16.6	14.9	14.1	12.6	12.2	11.3		8.0	6.4	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A1.3. Grenada: Sensitivity Analysis for Key Indicators of Public Debt, 2016-36

	Projections							
	2016	2017	2018	2019	2020	2021	2026	2036
PV of Debt-to-GDP Ratio								
Baseline	70	59	54	49	45	42	41	44
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	70	65	65	65	66	66	71	84
A2. Primary balance is unchanged from 2016	70	58	53	48	43	38	16	-17
A3. Permanently lower GDP growth 1/	70	60	56	52	49	47	55	87
A4. Alternative Scenario: Natural Disaster	70	64	58	54	50	47	46	48
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	70	64	65	62	59	58	66	84
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	70	65	66	61	56	53	52	51
B3. Combination of B1-B2 using one half standard deviation shocks	70	67	69	65	62	60	64	74
B4. One-time 30 percent real depreciation in 2017	70	78	72	67	62	59	54	53
B5. 10 percent of GDP increase in other debt-creating flows in 2017	70	67	61	57	52	49	48	48
PV of Debt-to-Revenue Ratio 2/								
Baseline	279	240	220	203	187	176	176	193
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	279	263	264	267	270	273	297	362
A2. Primary balance is unchanged from 2016	279	238	215	196	176	156	67	-77
A3. Permanently lower GDP growth 1/	279	243	226	213	201	194	231	381
A4. Alternative Scenario: Natural Disaster	279	258	238	221	206	194	195	213
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	279	258	260	249	240	236	277	369
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	279	266	269	249	233	221	219	227
B3. Combination of B1-B2 using one half standard deviation shocks	279	271	281	266	253	245	269	324
B4. One-time 30 percent real depreciation in 2017	279	318	296	276	258	243	228	235
B5. 10 percent of GDP increase in other debt-creating flows in 2017	279	273	250	232	216	205	204	215
Debt Service-to-Revenue Ratio 2/								
Baseline	85	71	66	64	61	59	47	34
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	85	72	68	67	66	65	58	54
A2. Primary balance is unchanged from 2016	85	71	66	63	61	58	44	18
A3. Permanently lower GDP growth 1/	85	72	67	65	63	62	52	48
A4. Alternative Scenario: Natural Disaster	85	73	68	66	63	61	49	36
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	85	75	74	72	69	67	55	47
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	85	71	67	65	62	60	50	37
B3. Combination of B1-B2 using one half standard deviation shocks	85	74	71	70	67	64	54	45
B4. One-time 30 percent real depreciation in 2017	85	75	74	71	68	66	53	42
B5. 10 percent of GDP increase in other debt-creating flows in 2017	85	71	67	65	62	60	49	36

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table A1.4. Grenada: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016-36

	Projections							
	2016	2017	2018	2019	2020	2021	**2026**	2036
PV of debt-to GDP ratio								
Baseline	46	38	34	31	28	26	**25**	26
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	46	44	45	46	48	49	**68**	92
A2. New public sector loans on less favorable terms in 2016-2036 2	46	38	35	33	30	28	**31**	39
A3. Alternative Scenario: Natural Disaster	46	42	38	35	32	30	**28**	27
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	46	40	39	36	32	30	**28**	30
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	46	39	39	36	33	31	**28**	27
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	46	39	36	33	30	28	**26**	28
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	46	41	40	37	34	32	**29**	27
B5. Combination of B1-B4 using one-half standard deviation shocks	46	43	45	42	38	36	**33**	31
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	46	54	48	44	40	37	**35**	37
PV of debt-to-exports ratio								
Baseline	162	137	122	112	101	93	**86**	86
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	162	158	160	166	170	174	**235**	298
A2. New public sector loans on less favorable terms in 2016-2036 2	162	138	126	117	107	101	**109**	125
A3. Alternative Scenario: Natural Disaster	162	151	137	126	115	107	**96**	87
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	162	137	122	112	101	93	**86**	86
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	162	153	163	150	137	128	**115**	103
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	162	137	122	112	101	93	**86**	86
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	162	147	144	133	122	114	**102**	89
B5. Combination of B1-B4 using one-half standard deviation shocks	162	151	158	146	134	125	**112**	98
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	162	137	122	112	101	93	**86**	86
PV of debt-to-revenue ratio								
Baseline	206	175	158	145	132	122	**117**	128
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	206	202	206	215	222	228	**320**	446
A2. New public sector loans on less favorable terms in 2016-2036 2	206	177	162	151	140	132	**148**	187
A3. Alternative Scenario: Natural Disaster	206	194	176	164	150	140	**131**	131
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	206	186	179	165	150	138	**133**	145
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	206	182	178	166	152	142	**134**	132
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	206	181	167	153	139	129	**124**	135
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	206	188	185	172	159	149	**138**	133
B5. Combination of B1-B4 using one-half standard deviation shocks	206	197	207	193	178	167	**156**	149
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	206	247	222	205	186	172	**165**	180

Table A1.4. Grenada: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016-36 (concluded)

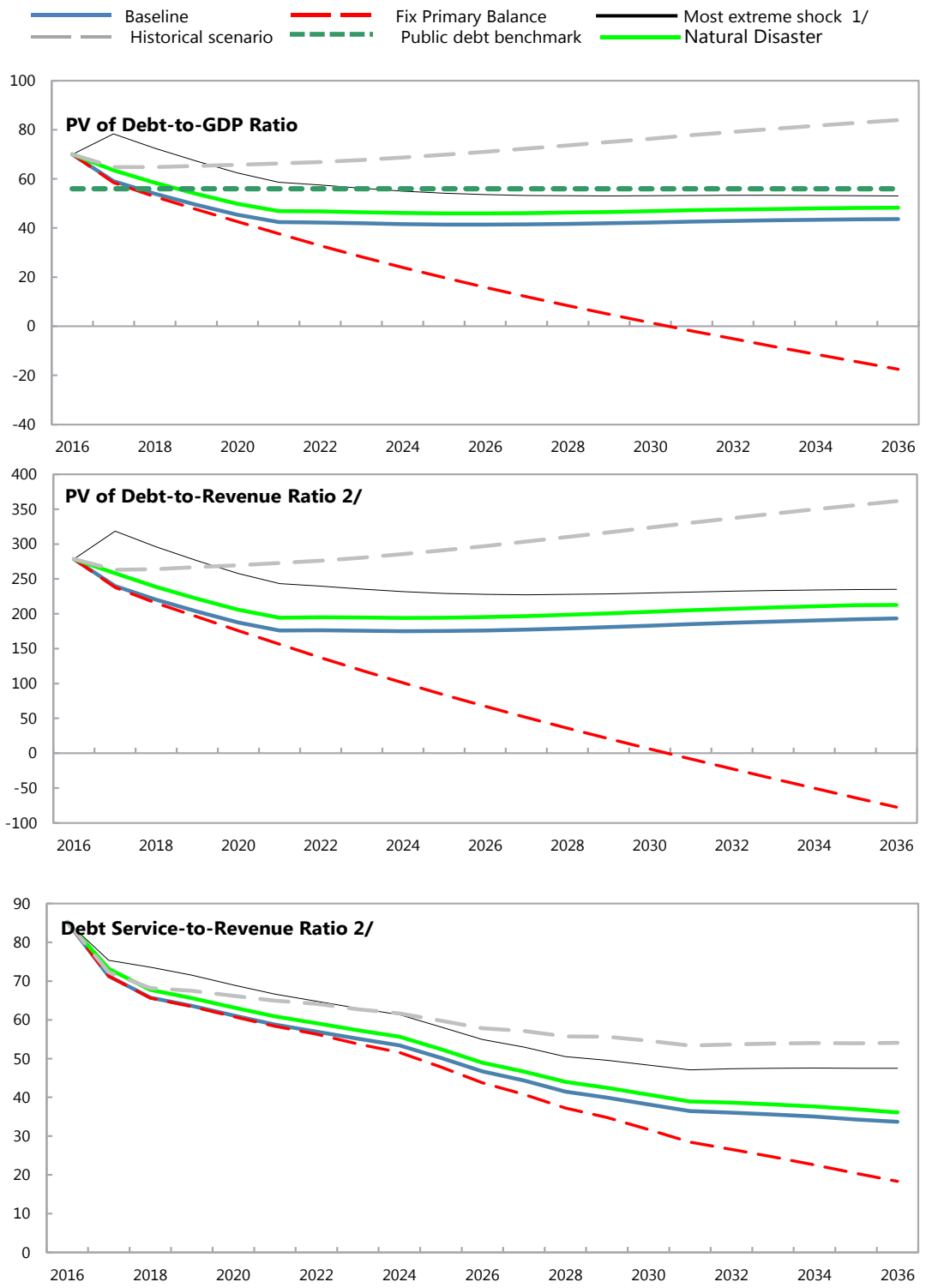
Baseline	18	17	16	14	14	13	**9**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	18	17	17	16	16	16	**18**	27
A2. New public sector loans on less favorable terms in 2016-2036 2	18	17	15	14	13	12	**10**	10
A3. Alternative Scenario: Natural Disaster	18	17	16	15	14	13	**10**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	18	17	16	14	14	13	**9**	7
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	18	18	19	17	17	16	**12**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	18	17	16	14	14	13	**9**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	18	17	16	15	14	13	**11**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	18	17	18	16	16	15	**12**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	18	17	16	14	14	13	**9**	7
Debt service-to-revenue ratio								
Baseline	23	21	20	18	18	17	**12**	11
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	23	22	22	21	21	21	**24**	40
A2. New public sector loans on less favorable terms in 2016-2036 2	23	21	20	18	17	15	**13**	15
A3. Alternative Scenario: Natural Disaster	23	21	21	19	19	17	**14**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	23	23	23	21	20	19	**14**	12
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	23	21	20	19	18	17	**14**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	23	22	21	19	19	18	**13**	11
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	23	21	21	19	19	17	**15**	12
B5. Combination of B1-B4 using one-half standard deviation shocks	23	23	23	21	21	20	**16**	14
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	23	30	28	26	25	24	**17**	15
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	28	28	28	28	28	28	**28**	28

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

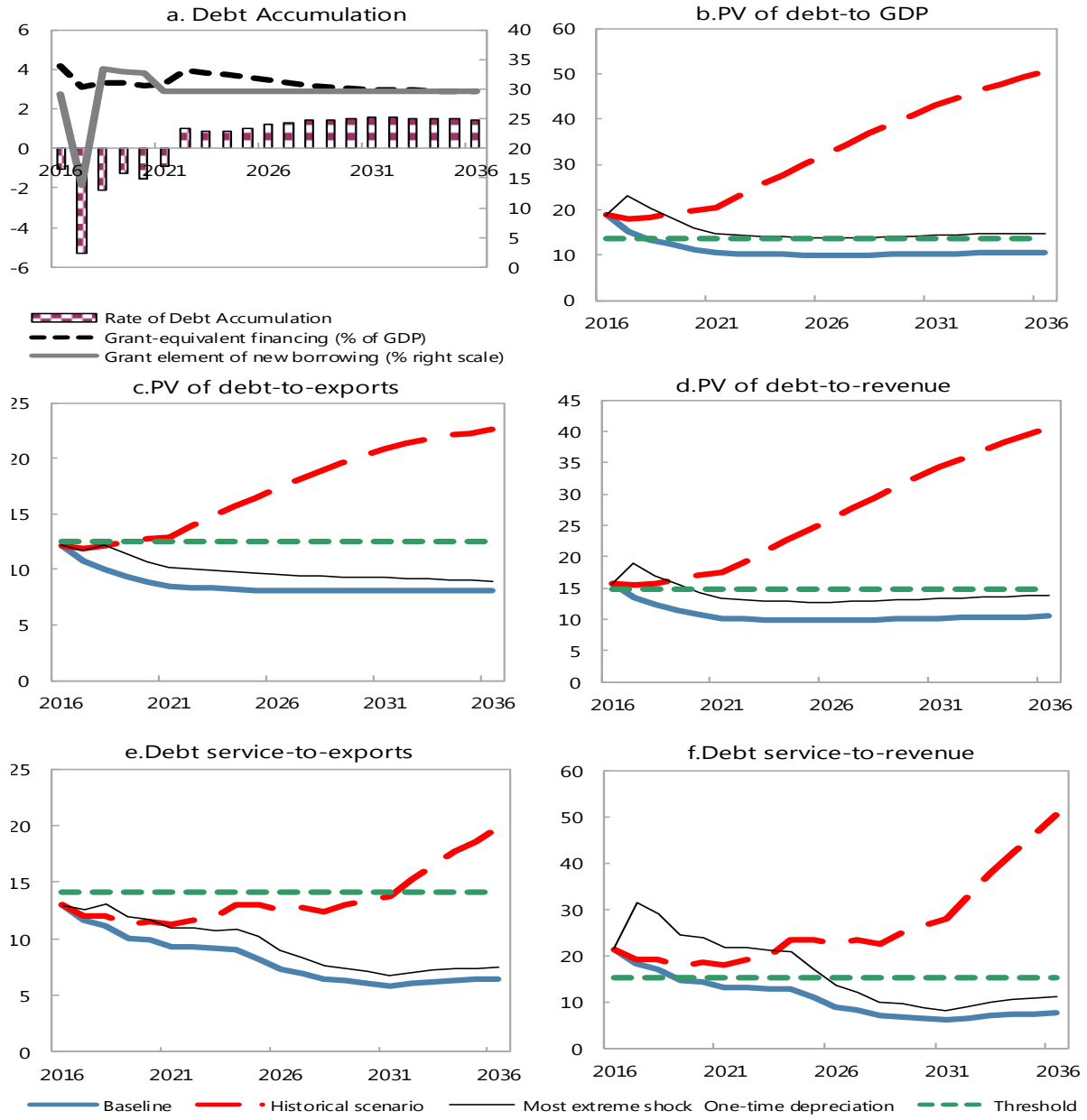
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming and offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure A1.1. Grenada: Indicators of Public Debt under Alternative Scenarios, 2016-36 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2026.
2/ Revenues are defined inclusive of grants.

Figure A1.2. Grenada: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2016-36 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2026. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock